NOETHERIANNESS ASCENDS AND DESCENDS ALONG A REPRESENTABLE SURJECTIVE FLAT FINITE TYPE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A Noetherian affine formal algebraic space is adic*, and the adic* property ascends and descends along such a morphism. Same chapter, Lemma lemma-implications-between-types at proof lines 4661-4663 and Lemma lemma-iff-adic-star at proof lines 4663-4665. Reduce to the case where both spaces are adic*.
- The criterion for Noetherianness of an affine formal algebraic space. Same chapter, Lemma lemma-characterize-noetherian-affine, invoked at proof lines 4665-4667. Reduces Noetherianness to a property of the stages of a presentation.
- A countable presentation of an affine formal algebraic space, and affineness of the base changes. Same chapter, Lemma lemma-countable-affine-formal-algebraic-space at proof lines 4667-4668 and Lemma lemma-affine-representable-by-algebraic-spaces at proof lines 4670-4672. Supply the stages and identify the base changes as affine schemes.
- Noetherianness ascends and descends along a faithfully flat finite type ring map. Algebra, Lemma algebra-lemma-descent-Noetherian and Lemma algebra-lemma-Noetherian-permanence, invoked at proof lines 4677-4679. Give the two directions of the conclusion at the level of stages.

Source reference. Original lemma 00FN: sources/stacks-chow/tags.txt, source lines 577–577. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 033E: sources/stacks-chow/tags.txt, source lines 4053–4053. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKM: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2558–2558. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AII: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2423–2423. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4642–4642. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIK: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2453–2453. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of affine formal algebraic spaces which is representable by algebraic spaces, surjective, flat, and (locally) of finite type. Then X is Noetherian if and only if Y is Noetherian.*

Proof. Observe that a Noetherian affine formal algebraic space is adic*, see Lemma 0AIK. Thus by Lemma 0AKT we may assume that both X and Y are adic*. We will use the criterion of Lemma 0AKM to see that the lemma holds. Namely, write $Y = \operatorname{colim} Y_n$ as in Lemma 0AII. For each n set $X_n = Y_n \times_Y X$. Then X_n is an affine scheme (Lemma 0AKN) and $X = \operatorname{colim} X_n$. Each of the morphisms $X_n \to Y_n$ is faithfully flat and of finite type. Thus the lemma follows from the fact that in this situation X_n is Noetherian if and only if Y_n is Noetherian, see Algebra, Lemma 033E (to go down) and Algebra, Lemma 00FN (to go up). □